# General Mathematics — Exercise Sheet 7

*Continuous functions and the derivative*

**Variant: Statements**

**Exercise 1 (Extreme values without differentiating).** Let $f(x) = x/(1+x^2)$ on $[0, 2]$. Explain why Lecture 7's extreme value theorem applies. Find both extreme values and every input attaining them by algebra alone. Decide whether the restriction to $(0, 1)$ attains either an infimum or a supremum. Use Lecture 5's continuity and limit laws.

**Exercise 2 (An inverse on a domain with a gap).** On $A = [0, 1) \cup \{2\}$ define $f(x) = x$ for $0 \le x < 1$ and $f(2) = 1$. Prove that $f : A \to [0, 1]$ is continuous relative to $A$, strictly increasing and bijective. Write its inverse and show that the inverse is not continuous at 1. Identify the missing hypothesis in Lecture 7's inverse theorem. Use the relative and sequential continuity tests from Lecture 5.

**Exercise 3 (Matching derivatives at a junction).** For $a, b \in \mathbb{R}$, define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} \sin x, & x < 0, \\ ax^2 + bx, & x \ge 0. \end{cases}$$

Determine all parameters for which $f$ is continuous at zero and all parameters for which it is differentiable there. Find the derivative when it exists, directly from one-sided quotients. Use Lecture 5's sine limit and Lecture 7's derivative definition.

**Exercise 4 (A differential with an exact error).** Let $f(x) = x^3$. Expand $f(2 + h)$ algebraically and use the linear approximation criterion to find $f'(2)$ without a derivative rule. Use the differential at 2 to approximate $f(2.02)$. Compute the exact error of the approximation and give a radius making the remainder at most $\varepsilon|h|$ for any $\varepsilon > 0$.